NORTHSTAR DISTRIBUTION: DEMAND, INVENTORY AND TRANSPORTATION ANALYSIS

Executive Summary

NorthStar Distribution Co. is an FMCG distributor supplying everyday products to 100 retailers across three US regions. To run efficiently, it must meet customer demand without tying up too much money in stock or spending unnecessarily on transportation. Weak alignment between demand and inventory leads to excess stock in some products and stockouts in others, while poor load planning leaves trucks partly empty and raises fuel and running costs. This project will analyze NorthStar's supply chain performance from 2022 to 2024 using Diagnostic, Optimization and Predictive analytics. It will identify operational inefficiencies, estimate their financial impact, forecast demand for Q1 2025, and recommend practical improvements.

Business Context & Stakeholders

The analysis is intended to support supply chain managers, inventory planners and transportation managers at NorthStar Distribution. It gives them a clear view of how demand is changing, where inventory is out of line with demand, how much working capital is tied up in excess stock, and where truck capacity and fuel are being lost. Supply chain managers can use it to prioritize improvements, inventory planners to adjust purchasing and reorder levels, and transportation managers to plan load consolidation and track delivery times.

Primary Objectives & Success Measures

The project aims to understand how demand has changed and what to expect next quarter; identify where inventory is out of line with demand and how much cash can be released safely; identify where fuel and truck capacity are being lost and which routes offer the biggest efficiency opportunity; and understand what is associated with late deliveries. Success will be measured using relevant operational KPIs, including Total Revenue, Total Units Sold, Forecast Q1 2025, Fuel Purchased (Gallons), Estimated Savings, Inventory Value, Excess Inventory, Potential Cash Freed, Inventory Days, Stockout Rate (%), Truck Utilization (%), Unused Truck Capacity (%), Fuel Cost on Empty Space, Average Delivery Days and Delayed Deliveries (%).

Key Business Questions

1. How has demand changed over time?
2. Which products and locations drive the biggest changes (2024 vs 2023, gains and drops)?
3. What is the forecast for next quarter (Q1 2025) by category and region, ranked by growth vs Q1 2024?
4. Where is inventory out of line with demand (stockout vs excess risk, latest week)?
5. How much working capital is tied up in excess inventory, and what cash can be released safely?
6. Where are we losing fuel and truck capacity?
7. What is associated with late deliveries (late = average delivery days of 4 or more)?
8. Which routes and activities (region × category) offer the biggest efficiency opportunity?
9. Which destination regions have the lowest truck utilization?
10. Which destination regions have the most loads under 60% utilization?

Data Sources, Privacy & Assumptions

The project will use the NorthStar Distribution FMCG Supply Chain Dataset from Kaggle as the primary data source, covering January 2022 to December 2024. The structured, transactional data includes approximately 190,000 daily sales records, 41,580 weekly inventory records, 25,729 supplier orders, and 88,158 truck-load lines representing 41,959 loads, alongside product, retailer, supplier, distribution center and fuel price details. Data will be cleaned and analyzed in SQL Server and presented in a three-page Power BI dashboard. The dataset is treated as a business/synthetic analytics dataset, with no personally identifiable information intentionally exposed.

Key assumptions and limitations will be documented. Transport costs are modelled using an assumed 45,000 lb truck payload, 6.5 miles per gallon, 600 miles per load and $1.60 per mile in other running costs, because distance and fuel per load are unavailable. Inventory is valued at 70% of selling price. Deliveries taking four or more days are treated as delayed because no promised delivery dates exist. Inventory records cover only 6 of the 50 facilities, and 2022 is treated as a start-up year and not used as a benchmark. Forecasts and potential savings will be presented as analytical estimates, not guaranteed outcomes.